

PEACE RIVER, AB, Canada

WMO: 710680

Lat: 56.230N

Lon: 117.450W

Elev: 1873

StdP: 13.73

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-30.0	-23.3	-38.3	0.7	-29.8	-31.6	1.0	-23.0	24.5	36.3	21.8	31.5	3.6	220	0.694

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.3	81.4	62.6	78.2	60.9	75.1	59.2	64.9	77.7	62.9	74.6	61.1	72.1	9.2	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
60.2	83.4	70.0	58.2	77.5	67.6	56.3	72.4	65.9	30.9	78.5	29.3	74.5	28.0	72.4	72.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 20.8	18.5	16.8	DB	-36.0	86.8	7.5	3.1	-41.4	89.0	-45.8	90.8	-50.0	92.6	-55.5	94.8	(4)
(5)			WB	-36.2	68.0	7.4	2.6	-41.5	69.9	-45.8	71.4	-50.0	72.9	-55.4	74.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	35.0	5.8	10.5	20.8	37.6	50.3	58.0	61.6	59.1	50.1	37.4	19.3	8.4		
	DBStd	23.05	17.39	14.55	14.01	9.36	7.98	5.55	4.97	6.03	7.55	8.48	12.70	15.29		
	HDD50	6561	1371	1106	906	378	94	6	1	5	89	396	920	1289		
	HDD65	10984	1836	1526	1371	821	458	219	128	199	447	855	1370	1754		
	CDD50	1102	0	0	0	7	103	245	361	287	92	7	0	0		
	CDD65	49	0	0	0	0	1	8	23	16	1	0	0	0		
	CDH74	898	0	0	0	4	60	166	354	272	42	0	0	0		
	CDH80	157	0	0	0	0	9	22	61	61	4	0	0	0		
	WSAvg	7.2	6.6	6.6	7.3	8.2	8.4	7.6	7.0	6.9	7.4	7.5	6.7	6.4		
	PrecAvg	16.2	0.9	0.9	0.6	0.7	1.3	2.6	2.5	2.2	1.7	0.9	1.0	0.8		
	PrecMax	26.0	2.4	2.6	1.5	2.4	4.8	5.9	5.6	4.8	5.9	2.0	1.9	2.3		
(15) Precipitation	PrecMin	8.3	0.2	0.0	0.0	0.0	0.1	0.4	0.6	0.3	0.4	0.1	0.0	0.2		
	PrecStd	3.5	0.6	0.7	0.4	0.5	1.1	1.6	1.3	1.2	1.2	0.5	0.6	0.5		
	DB	43.1	41.9	50.3	70.3	80.1	83.1	85.5	86.0	79.2	68.1	50.0	43.9			
	MCWB	37.0	35.9	41.2	50.7	57.3	61.7	65.7	64.6	60.8	53.7	42.2	37.4			
	2% DB	39.4	38.5	45.0	62.3	74.4	78.5	81.2	80.8	73.6	59.1	43.0	39.0			
(23) Mean Coincident Wet Bulb Temperatures	MCWB	34.6	33.4	37.3	46.3	54.3	59.4	63.5	63.5	58.0	47.8	37.1	34.1			
	5% DB	35.2	35.1	41.4	57.3	70.3	75.1	78.1	76.5	68.8	54.3	38.9	34.9			
	MCWB	31.5	30.8	35.0	43.6	52.5	57.8	62.2	61.1	55.5	44.5	34.5	31.0			
	10% DB	30.6	30.9	38.7	53.0	66.4	71.5	74.6	72.8	64.6	50.7	35.3	30.2			
	MCWB	27.3	27.3	33.3	41.5	50.5	56.5	60.5	59.2	53.1	42.6	31.8	27.5			
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4% WB	37.4	36.5	41.4	51.4	58.8	64.5	68.4	67.1	62.5	54.3	42.4	37.7			
	MCDB	42.5	40.9	49.5	68.2	75.5	78.4	80.6	80.7	75.9	67.0	49.4	43.7			
	2% WB	34.8	33.8	37.8	47.2	56.0	61.5	65.6	64.6	59.2	48.7	38.0	34.2			
	MCDB	39.1	38.1	43.8	61.1	71.0	74.3	77.5	77.9	71.7	57.2	42.4	38.6			
	5% WB	31.8	30.9	35.6	44.4	53.6	59.5	63.5	62.4	56.6	45.6	34.7	31.0			
(31) Mean Daily Temperature Range	MCDB	35.3	35.0	40.8	55.7	67.9	72.0	75.1	73.9	67.2	52.6	38.4	34.5			
	10% WB	27.2	27.3	33.4	41.9	51.5	57.5	61.7	60.3	53.9	43.1	32.1	27.4			
	MCDB	30.3	30.9	38.3	52.0	64.6	68.9	72.3	70.6	63.0	50.1	34.9	30.2			
	5% DB	15.8	17.7	18.4	19.8	22.6	20.7	21.3	22.2	21.6	17.3	14.2	15.4			
	MCDBR	19.4	20.2	19.5	28.2	29.5	27.8	28.0	28.8	29.3	24.0	17.4	18.6			
(36) Clear-Sky Solar Irradiance	MCWBR	16.0	16.5	13.9	16.5	15.2	13.4	13.6	14.2	16.2	15.2	13.4	15.4			
	5% WB	19.1	19.9	18.8	26.0	27.1	24.0	25.2	25.8	28.1	22.6	16.7	18.2			
	MCDBR	16.0	16.5	13.6	15.7	14.2	12.4	13.3	13.7	16.1	15.0	13.3	15.4			
	taub	0.241	0.247	0.291	0.327	0.340	0.352	0.353	0.349	0.311	0.283	0.242	0.224			
	taud	2.258	2.341	2.310	2.374	2.409	2.397	2.414	2.409	2.525	2.558	2.418	2.218			
(42) All-Sky Solar Radiation	Ebn at Noon	218	269	279	284	285	282	280	273	272	254	223	194			
	Edh at Noon	16	22	30	33	34	35	33	31	24	18	14	13			
	RadAvg	222	532	1025	1483	1781	1896	1847	1497	1025	545	239	151			
(45)	RadStd	22	48	86	96	136	152	119	126	81	69	21	11			

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (20 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+103	N/A	

Nomenclature: See separate page